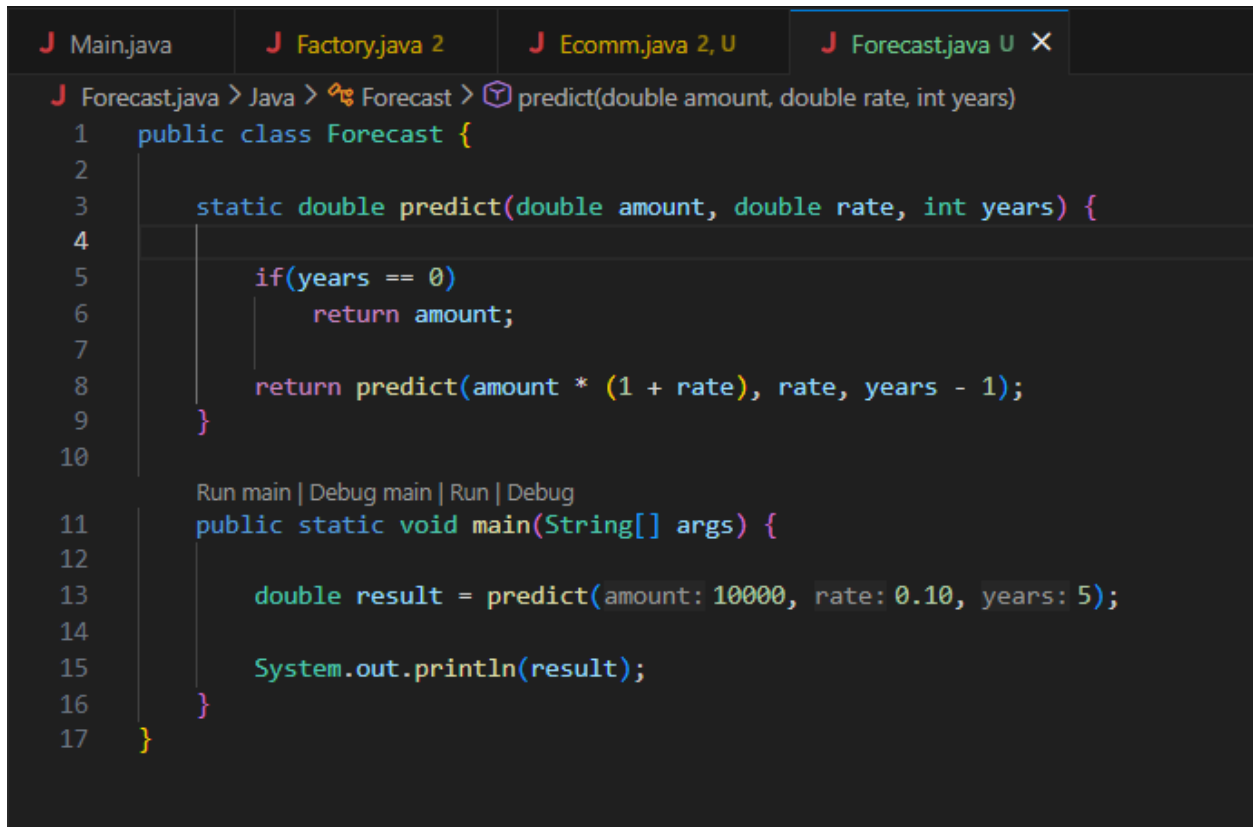


Week 1

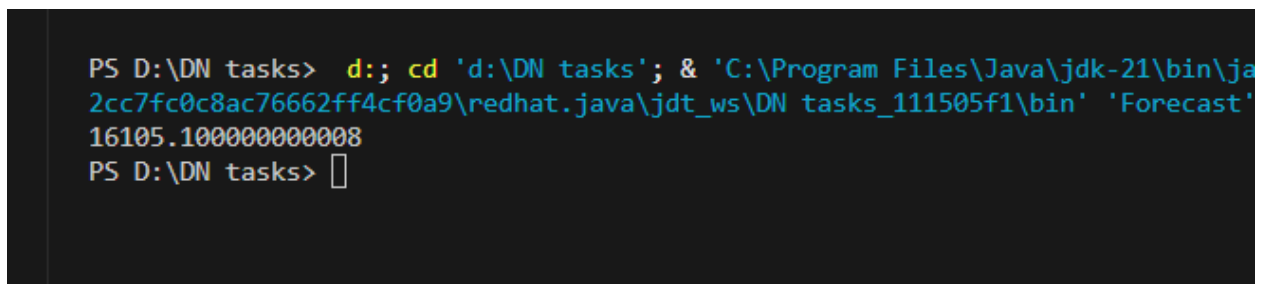
Exercise 4: Financial Forecasting

Code:



```
J Main.java J Factory.java 2 J Ecomm.java 2, U J Forecast.java U X
J Forecast.java > Java > Forecast > predict(double amount, double rate, int years)
1 public class Forecast {
2
3     static double predict(double amount, double rate, int years) {
4
5         if(years == 0)
6             return amount;
7
8         return predict(amount * (1 + rate), rate, years - 1);
9     }
10
11     Run main | Debug main | Run | Debug
12     public static void main(String[] args) {
13
14         double result = predict(amount: 10000, rate: 0.10, years: 5);
15
16         System.out.println(result);
17     }
```

Output:



```
PS D:\DN tasks> d:; cd 'd:\DN tasks'; & 'C:\Program Files\Java\jdk-21\bin\java.exe' -cp 'C:\Program Files\Java\jdk-21\bin\java\2cc7fc0c8ac76662ff4cf0a9\redhat.java\jdt_ws\DN tasks_111505f1\bin' 'Forecast'
16105.1000000000008
PS D:\DN tasks> 
```